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Optimizing Well Construction: Benefits of Integrating the Metal Expandable Packer with a Stage Collar

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Abstract

This paper explores the integration of a metal expandable packer with a stage collar as a solution to address challenges in two-stage cementing operations. By leveraging the reliability and adaptability of metal expandable packers, along with the operational flexibility of decoupled stage collars, this approach aims to mitigate the limitations inherent in traditional DV tools. The proposed system not only enhances zonal isolation but also optimizes cement placement efficiency, particularly in complex wellbore geometries or high-pressure environments.

The integration of the metal expandable packer with a stage collar creates a robust alternative to traditional methods, addressing issues such as mechanical complexity and limited zonal isolation. Unlike conventional inflatable packers, the metal expandable packer ensures a more reliable seal, even in irregular or washed-out boreholes, thanks to its ability to conform precisely to the wellbore geometry. Furthermore, the decoupling of the packer and stage collar introduces a new level of operational adaptability, allowing the use of these components independently based on real-time well data and specific cementing requirements.

This novel approach also proves advantageous in extreme well environments, where high-pressure, high-temperature (HPHT) conditions or chemically aggressive fluids are present. The packer's construction from advanced alloys enhances its resistance to deformation, thermal expansion, and corrosion, ensuring durability over the life of the well. Additionally, the system simplifies operational logistics by reducing the need for complex mechanical assemblies, thereby lowering the risk of tool failure or operational delays.

Deployed on 13-3/8-in. casing within 18-5/8-in. casing, the packer features a slim profile and a robust multi-element sealing system, enabling high-rate circulation, full rotation, and improved mud displacement. The tool was qualified to ISO 14310 V0 standards and validated through extensive testing cycles. During field deployment, a contingency procedure was executed when the primary cementing plug failed to land; a free-fall backup plug enabled successful setting of the packer. Thus, eliminating static losses and confirming effective annular isolation. The paper outlines the system's design, testing, and installation process, and discusses its potential for adaptation to various casing sizes and multi-stage cementing applications.

Introduction

Cementing is essential in well construction for zonal isolation, casing support, and protecting the wellbore from formation fluids and pressure regimes. In deep or complex wells, achieving effective cement coverage in a single stage operation can be challenging due to factors such as long casing strings, narrow annular clearances which can result in induced loss of circulation while cementing, pressure limitations at surface and cementing across loss-prone formations. In these instances, the application of two-stage cementing is a well-established method to improve cement placement and achieve comprehensive annular coverage.

Two-stage cementing involves placing cement in two separate intervals—typically, a first stage through the casing shoe to isolate deeper zones, followed by a second stage through a stage collar positioned higher on the casing to ensure coverage in the upper annulus. This approach reduces hydrostatic pressure during cement placement, minimizes losses in weak formations, and improves displacement efficiency.

Traditional two-stage cementing operations typically rely on DV (Differential Valve) tools integrated within the casing string to facilitate the second-stage cement job. While these tools have been widely used and proven in various field applications, they present several inherent limitations, particularly in demanding or high-integrity well environments.

Conventional DV tools are limited by their mechanical complexity and reliance on internal sleeves and shear mechanisms, which are sensitive to debris, temperature, and pressure changes. This can cause inconsistent valve port operation, an increase in the risk of cement placement failures and thus impacting zonal isolation. Another disadvantage of conventional DV tools is the reduction in annular clearance due to their outer diameter versus the inner diameter of the existing casing, leading to higher pressure regimes during cement slurry displacement.

Furthermore, DV tools are often integrated into a fixed position within the casing string, which may limit flexibility in stage collar placement. This fixed geometry may not align with optimal cementing zones, particularly in wells with variable formation pressures or where specific isolation intervals are needed. Repositioning the DV tool may require a complete redesign and remanufacture of the casing string. Another critical drawback of conventional systems is the limited annular isolation capability, especially in large-diameter casing strings or irregular boreholes. Inflatable packers used alongside DV tools often suffer from inconsistent expansion or sealing failures due to borehole washouts or non-uniform contact.

The metal expandable packer with a decoupled stage collar offers several key advantages:

- Independent placements of the packer and stage collar, allowing precise customization of cementing stages based on actual well conditions.
- High-pressure, high-temperature (HPHT) rated alloys offer enhanced long-term reliability and improved resistance to differential pressures and exposure to harsh wellbore fluids.
- Slim design and high-flow capacity, enabling efficient mud displacement and full casing rotation during deployment, which improves cement bonding and job execution.

This innovative solution addresses the mechanical and operational limitations of conventional DV tool systems, thereby improving cementing outcomes, well integrity, and adaptability to diverse wellbore conditions.

Metal Expandable Packer Design

The development and deployment of the hydraulically set metal expandable packer with a separately located stage collar followed a structured engineering and qualification process. The system was designed to be run on a 13-3/8-in 86# VM-110 HCSS ([Figure 1](#)) casing string and set inside an existing 18-5/8-in 136# L-80 casing, with the packer and stage collar functioning as independent units to allow optimized slurry placement based on well design and cementing requirements.



Figure 1—Metal Expandable Packer Mounted on 13-3/8-in 86# VM-110 HCSS

The host casing's internal diameter of 17.239 inches presents significant run-in-hole risks for conventional DV tools due to their large outer diameter (maximum OD of 16.875 inches). In contrast, the metal expandable packer, with its slim 15.5-inch OD, can be deployed smoothly within the casing without the need for a dedicated string mill run—eliminating additional operational time and reducing deployment complexity.

The metal expandable packer featured a 32-inch HNBR/PTFE seal stack. Serving as multi-element sealing system with built-in redundancy. This configuration allows for effective sealing across a wide expansion range while maintaining the ability to rotate the casing and circulate at high flow rates during deployment. Material selection and mechanical design were optimized for pressure containment, expansion consistency, and corrosion resistance in downhole environments.

The MEP complies with the top industry packer standard ISO 14310 V0 (V-Zero) for cased hole applications, featuring a differential pressure rating of 4000 psi both above and below the packer. It also includes the capability to restore casing integrity after setting, maintaining a burst pressure of 9000 psi and a collapse pressure of 6560 psi, consistent with the casing specifications.

Job Preparation

The effective implementation of the metal expandable packer and the stage collar system necessitated comprehensive job preparation and a compatibility assessment among all tools, plugs, and pressure activation systems.

A major challenge during job design was the exceptionally large first-stage cement volume, which exceeded the offshore rig's capacity to handle and store mixing chemicals in the limited available space. As a result, it was necessary to position the packer as deep as possible. However, placing the packer deeper also extended the free-fall time required for the opening plug. In this context, the combined Metal Expandable Packer (MEP) and stage collar technology provided a significant advantage: by decoupling the stage collar from the packer, it was possible to position the stage collar around 2,000 ft above the MEP, optimizing both logistics and plug deployment time. See [Figure 2](#) for details.

Operational contingencies, including the potential failure to land the cement shut-off plug, were considered and a plan was developed to include a contingency free fall landing collar positioned below the MEP, as illustrated in [figure 2](#). This is necessary if the shut-off plug does not bump during the first stage cementing operation. The Metal Expandable Packer requires a closed system to avail the ability to pressure up to the setting pressure for the packers.

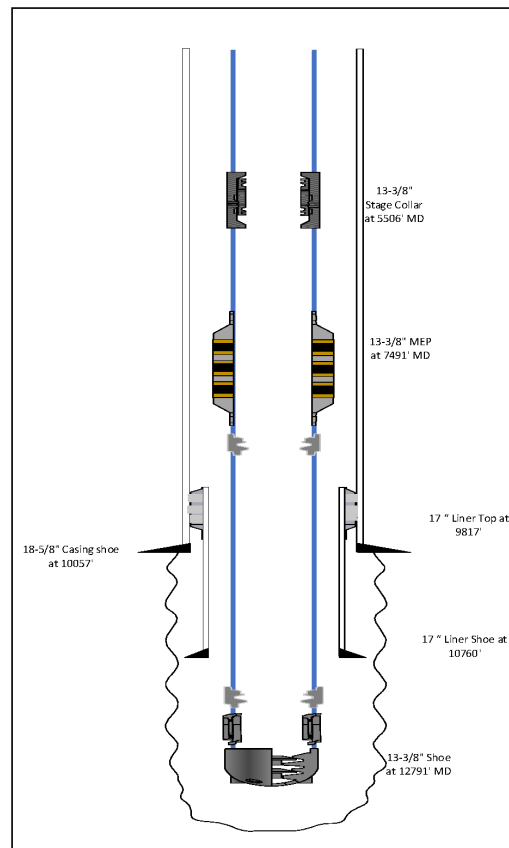


Figure 2—Well schematic illustrating the spatial separation and offset distance between the metal expandable packer and the stage collar.

In this two-stage cementing system, it was necessary to drop a total of 5 plugs, as shown in [figure 3](#):

1. Bypass plug, utilized primarily to prevent contamination before the cement.
2. Shut-off plug, used to displace the first stage cement and set the MEP by isolating the pressure.
3. Opening free fall plug, designed to shift the stage collar open for the second stage.
4. Closing plug, employed to displace the second stage cement and shift to close the stage collar.
5. Contingency free fall plug, an emergency option in the event the first stage shut-off plug does not bump.

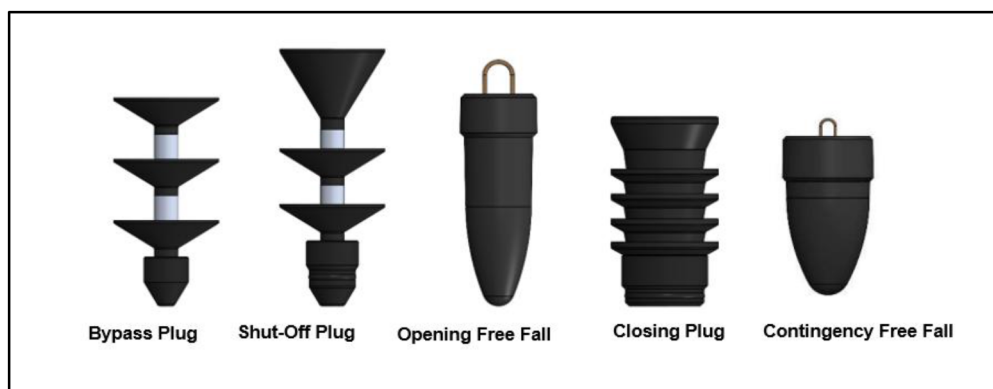


Figure 3—Cementing Plugs

A thorough pre-job inspection was conducted to verify the dimensions and compatibility of all plugs with the stage tool and contingency landing collar. Personnel were trained on the unique handling and running procedures and were briefed on rig-floor interface requirements for activation and confirmation of the setting sequence.

One of the critical aspects of the job design was to model the cementing operation to accurately predict the maximum surface pumping pressure during the first-stage cement job. This was essential to ensure that the Metal Expandable Packer (MEP) would not prematurely expand, particularly during cement lift. As shown in the [figure 4](#) below, the maximum calculated surface pressure was 1,240 psi, which remains safely below the packer's expansion yield threshold.

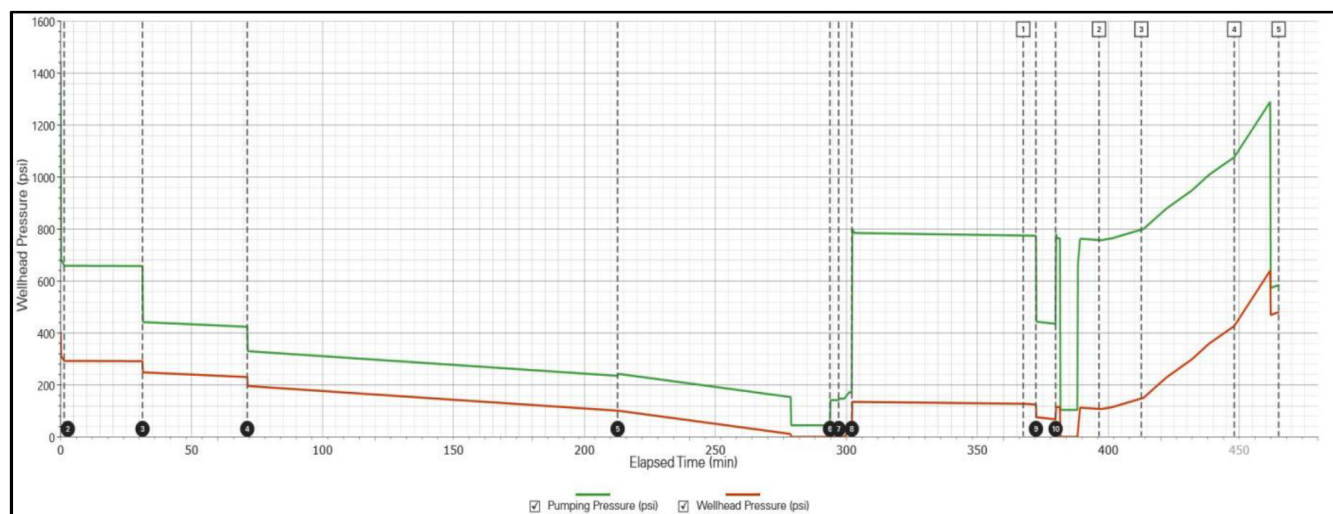


Figure 4—Surface pressure

Cementing and Metal Expandable Packer (MEP) Expansion

The 16-inch hole section was drilled in a vertical orientation utilizing a mud weight of 15.7 pounds per gallon (ppg), successfully reaching a total measured depth of 12,800 feet, referenced from the Rotary Kelly Bushing (RKB). The previous casing (expandable liner) shoe was set at 10,760 feet. During the drilling phase of the 16-inch hole, the first half of the open hole was drilled with full circulation. However, as drilling operation approached the casing point, dynamic fluid losses began to occur. These losses gradually increased in severity, eventually reaching an estimated rate of approximately 120 barrels per hour, suggesting the presence of a highly permeable or fractured zone about 1,000 ft above the planned casing depth. This added to the challenges of drilling across the bottom section known for highly pressured zones while ensuring loss of circulation is reduced to the barest minimum.

The 13-3/8-inch casing point was strategically chosen to be set at the top of a Gas Reservoir formation to provide zonal isolation and mechanical support.

The 13-3/8-inch casing was successfully deployed to bottom and cemented using a two-stage cement slurry design. The first-stage cement operation began with the placement of the bottom wiper plug, followed by the mixing and pumping of 423 barrels of 16.7 ppg blended lead slurry. Subsequently, 187 barrels of 17.4 ppg tail slurry were blended and pumped (see [Figure 5](#)), after which the top plug was released.

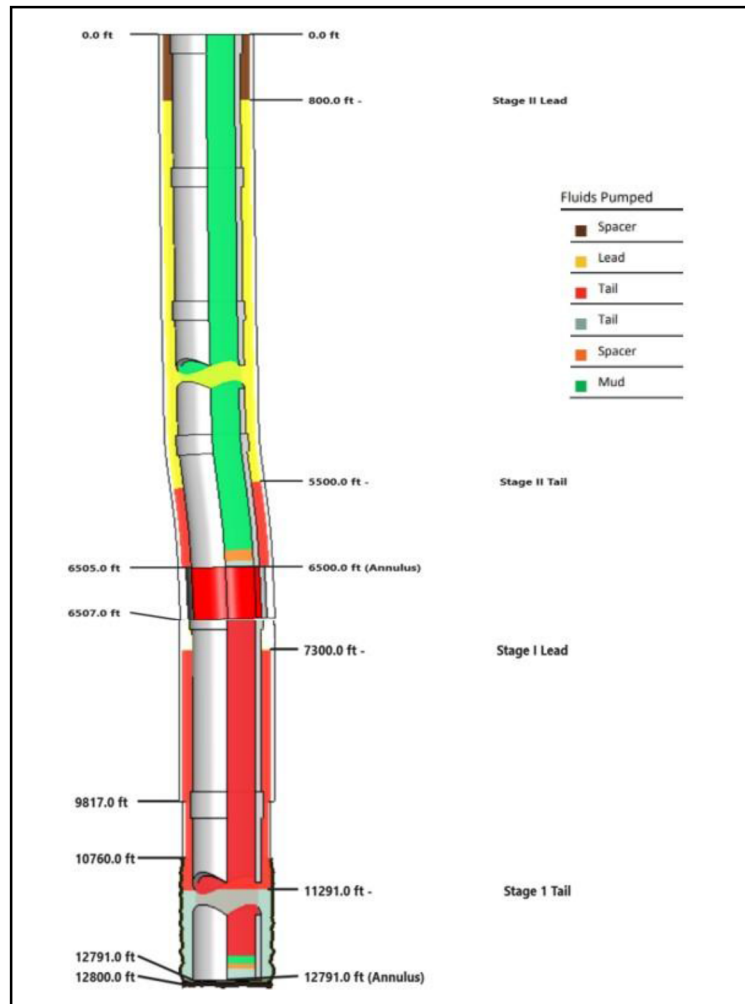


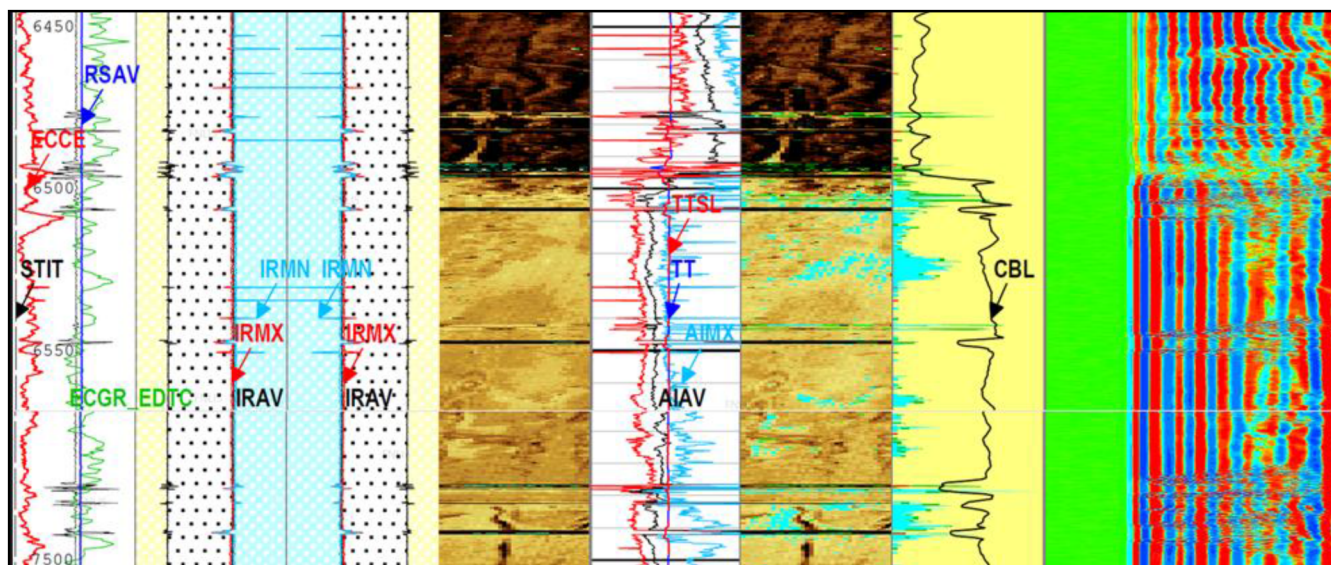
Figure 5—Fluid Positions

Rig pumps then displaced the staged cement at rates of up to 12 bpm while maintaining 62 percent returns throughout the displacement, circulating 822 barrels of 15.7 ppg water base mud (WBM), 60 barrels of 16.3 ppg tuned spacer, and 897 barrels of 15.7 ppg WBM. Pumping was slightly reduced as the plug passed the stage collar and the contingency landing collar. Although the plug did not bump, surface pressure readings indicated a successful cement lift. The loss of circulation rate while pumping and displacing the first stage slurries reduced from as high as 150 barrels per hour at the initial stages to 60 barrels per hour at the end of the displacement operation. Since the shut-off plug did not land, it was necessary to deploy the contingency free-fall activation dart to allow it to land on the contingency collar and to subsequently enable setting of the well annular barrier, MEP, packer.

The annulus was monitored via the trip tank, which showed a static loss rate of 40 barrels per hour. Finally, the MEP was expanded using the cement unit: pressure was gradually increased to 1,000 psi, then raised in 500 psi increments up to 3,200 psi (held for 10 minutes with no static losses), and subsequently increased to 3,750 psi for five minutes—again confirming zero static losses.

Results

The quality of cement placement was evaluated using downhole logging tools and surface monitoring data. Cement bond logs (CBLs) and variable density logs (VDLs) were acquired across the 13-3/8" casing interval to quantify zonal isolation and verify the integrity of the annular cement sheath see [Figure 6](#).



At depth 7491 ft, the cement bond is very good based on multiple indicators from the log. The CBL amplitude indicates strong signal attenuation, which corresponds to a solid bond between the casing and the formation. The bond type image at this depth confirms effective cement placement and good zonal isolation. Additionally, the Sonic VDL shows faint and attenuated waveforms with no strong first arrivals, which is consistent with a well-bonded casing. The acoustic impedance and transit time logs display smooth, stable curves without anomalies, further supporting the conclusion that the casing is well-cemented at this interval.

In summary, the integration of downhole logging results (CBL, VDL, sonic, impedance) and surface monitoring data conclusively demonstrated that:

- The cement sheath provided exceptional bond quality above the packer, with an average CBL amplitude of under 5 mV and a bond index of 0.95 in the 7,400–7,600 ft interval.
- Loss of circulation observed during the first stage cementing operation and the static loss rate redorded immediately afterwards completely stopped after the expansion of the MEP indicating the packer's ability to isolate the annulus and restore wellbore integrity.

These results validate the performance of the Metal Expandable Packer and stage collar system achieving dependable well construction and ensuring cement integrity, even under challenging conditions involving high pressures and significant fluid losses.

Conclusion

The field deployment presented in this case study highlights the significant advantages achieved through the implementation of the novel system, integrating a hydraulically set Metal Expandable Packer (MEP) with a decoupled stage collar for two-stage cementing in challenging well environments. By separating the functions of the packer and stage collar, operators gain unprecedented flexibility to optimize stage collar placement depths, minimize free-fall times, and tailor cement volumes to meet specific downhole requirements — advantages that are not attainable with conventional dual-valve (DV) tool systems.

Key conclusions from this study include:

- ### 1. Enhanced well integrity

Rapid packer expansion at controlled pressures up to 3,750 psi without premature activation, ensuring an effective seal against the 18-5/8" host casing. Downhole acoustic logs (CBL/VDL)

confirmed that cement bond quality above the packer was outstanding, with amplitude readings under 5 mV and a bond index exceeding 0.90. Deeper intervals maintained acceptable bond indices of 0.85–0.90 despite the open hole losses, thereby verifying that the MEP, can deliver uniform cement coverage acting as a solid base for the second stage cement.

2. Operational Flexibility and Risk Mitigation

The decoupled stage collar concept allowed the second-stage port to be positioned approximately 2,000 ft above the packer, enabling optimized cement displacement in a high-loss zone and reducing the rig's logistical burden associated with large slurry volumes. When the primary shutoff plug failed to land on the landing collar, the free-fall contingency plug technique ensured successful pressurization of the packer, eliminating fluid losses after the first-stage cement job. In addition, the use of a multi-element HNBR/PTFE seal stack enabled redundancy therefore minimizing the risk of sealing integrity in the casing.

3. Rig Time Cost Savings

The slim outer diameter (OD) design of the MEP—measuring only 15.5 inches—enabled it to be run in hole without requiring an additional string mill run that would have been necessary to accommodate a conventional DV tool. This directly contributed to reduced rig time and overall cost savings. In addition It improved cement displacement efficiency.

4. Compatibility with Industry Standards

The MEP's ISO 14310 V0 qualification provided operator confidence in achieving casing integrity under both internal and external pressure regimes up to 4,000 psi against gas leaks. Pressure collapse and burst ratings (6,560 psi collapse, 9,000 psi burst) aligned with the 18-5/8" L-80 casing performance envelope, confirming that the packer could withstand subsequent completion operations without compromising mechanical integrity.

Although this case study focused on a 13-3/8"x18-5/8" casing configuration in a vertical well, the underlying principles of a slim, hydraulically actuated packer with a separate stage collar are readily transferable to other casing sizes, and multi-stage cementing scenarios "three stage cementing". Increased flexibility in collar placement, combined with robust packer sealing, positions this technology as a favourable alternative in offshore, deepwater, and high-pressure/high-temperature (HPHT) applications where reliability is paramount.

In conclusion, the integration of a Metal Expandable Packer with a decoupled stage collar represents a significant advancement in high-integrity well construction. By overcoming the mechanical limitations of DV tools, reducing rig time with eliminating extra string mill runs, and delivering good cement job under challenging downhole conditions, this system enhances well integrity and unlocks new opportunities for complex stage cementing.

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